

the function $\mathcal{Q}_{n_1, n_2}(k)$ which has expected density $\approx 1/N$ into roughly N many harmonics $\lambda_{\chi\xi^\ell}(k)$.

Invoking Conjecture $C_1(\varepsilon)$, we obtain for $\chi\xi^\ell \neq 1$

$$\sum_{k \asymp NX} \Upsilon_{n_1, n_2}(k) \lambda_{\chi\xi^\ell}(k) \ll X^{1/3+2\varepsilon/3},$$

which is sufficient provided that $N < X^{1/3-2\varepsilon/3}$. We merely require this bound on average over the complete family of $n_1, n_2 \sim N$, $\chi \in \widehat{\mathfrak{C}(d)}$, and $|\ell| \ll R_d$ (see Remark 7.1). It is tempting to think that this problem would be amenable to large sieve techniques but unfortunately, the cubic forms appearing in $\Upsilon_{n_1, n_2}(k)$ are entangled with n_1, n_2 . We mention that similar sums for a fixed discriminant and a fixed binary cubic or quartic form arise naturally in the context of Manin's conjecture for Châtelet surfaces [37].

The task is then to evaluate the main term coming from $\chi\xi^\ell = 1$

$$\sum_{n_1, n_2 \sim N} \frac{\beta_{n_1} \beta_{n_2}}{\sqrt{n_1 n_2} L(1, (\frac{4n_1 n_2}{\cdot}))} \sum_{k \asymp NX} \Upsilon_{n_1, n_2}(k) \lambda_1(k).$$

It is not obvious how to make the argument unconditional and we have to assume that $\lambda_1(k)$ may be replaced by the truncated approximation $\lambda_1^\sharp(k, T)$. For the divisor function along binary cubic forms Greaves has shown an asymptotic formula with a power saving [11].

Using Heath-Brown's large sieve for quadratic characters [17], on average over n_1, n_2 we can replace the factor $L(1, (\frac{4n_1 n_2}{\cdot}))^{-1}$ by $\sum_{k \leq X^\varepsilon} \frac{\mu(k)}{k} (\frac{4n_1 n_2}{k})$ (see Lemma 3.5). For the truncated approximation $\lambda_1^\sharp(k, T) = \sum_{c \leq T} (\frac{4n_1 n_2}{c})$ we can evaluate the sum over y_1, y_2 by applying Poisson summation twice, which produces a count for cubic congruences

$$(1.1) \quad \#\{(y_1, y_2) \in (\mathbb{Z}/c\mathbb{Z})^2 : n_1 y_2^3 \equiv n_2 y_1^3 \pmod{c}\}.$$

Expanding (1.1) by cubic Dirichlet characters, the principal characters give the main term. We then need to bound error terms of the form

$$\frac{1}{N} \sum_{c \leq T} \frac{1}{c} \sum_{k \leq X^\varepsilon} \frac{\mu(k)}{k} \sum_{\substack{\chi \pmod{c} \\ \chi^3 = \chi_0 \neq \chi_0}} \left| \sum_{n \sim N} \beta_n \chi(n) \left(\frac{n}{ck}\right) \right|^2.$$

This is bounded using the large sieve for sextic characters due to Baier and Young [1].

1.2.3. Type I_2 sums. The obtained Type I and Type II ranges would already be enough for a lower bound of the correct order of magnitude for the number of primes $p = x^2 + y^3$. To show an asymptotic formula, we also need to consider the Type I_2 sums. We are able to handle these for

$$D_2 < X^{1/4}$$

by using the spectral methods of $GL(2)$ automorphic forms. The proof will appear in an upcoming joint work with Grimmelt [14], as an application of the *averages over orbits* introduced in [13]. To sketch the idea in the critical case $d \sim D_2$, $m \sim n \sim \sqrt{X/D_2}$, note that for a fixed y we consider a variant of the divisor problem along a quadratic polynomial $\sum_x f(\frac{x}{A}) d(x^2 + y^3)$. Applying Poisson summation on x produces the expected main term and an error term with the dual variable of length $H \approx \sqrt{D_2}$.